# Kumar Abhinav

**AI Security Engineer | LLM & Agentic AI Security | SIEM Architecture**
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## PROFESSIONAL SUMMARY

Security engineer with ~2 years in a multi-client MSSP SOC and hands-on experience designing and securing an agentic AI system end to end. Architect and lead builder of Swaraj Hansa, an LLM-assisted multi-tenant SIEM platform at ESDS, implementing prompt-injection defences, structured-output fail-safes, human-in-the-loop review and four-layer tenant isolation, mapped to the OWASP Top 10 for LLM Applications. Background in L1–L2 incident response on QRadar, Seceon and Wazuh, plus web and mobile application security testing.

## SKILLS

**AI / LLM Security:** OWASP Top 10 for LLM Applications, prompt injection defence, LLM guardrails, Model Context Protocol (MCP) tool scoping, excessive-agency controls, structured output validation, human-in-the-loop review, MITRE ATLAS, NIST AI RMF

**AI Engineering:** Agentic AI architecture, local LLM deployment (Ollama, Open WebUI), prompt engineering, Python, FastAPI, AI-assisted development

**Security Architecture:** Multi-tenant isolation, Zero Trust, mTLS / PKI, IPsec, PostgreSQL row-level security (RLS), least privilege, threat modelling, Microsoft MCRA & MCSB, ISO/IEC 27001

**SIEM & Detection:** Wazuh, IBM QRadar, Seceon, OpenSearch, Logstash, Apache Kafka, MITRE ATT&CK, threat intelligence enrichment (AbuseIPDB, VirusTotal), incident triage

**AppSec & Platform:** Burp Suite, Nmap, Metasploit, Nessus, OWASP Top 10, PTES, Docker, Linux, Ansible, PostgreSQL, Redis

## PROFESSIONAL EXPERIENCE

**Security Analyst | ESDS Software Solution Limited** **Bengaluru | Nov 2024 – Present**

- Architected and led the build of Swaraj Hansa, an agentic multi-tenant SIEM for MSSP clients (Wazuh → Logstash → Kafka → normaliser → AI triage → PostgreSQL + OpenSearch → FastAPI/React), using AI coding agents under a self-authored architecture and security design.
- Hardened the LLM triage path against prompt injection: sanitised untrusted alert fields (control-character stripping, injection-phrase neutralisation, 2,000-character truncation) and isolated log data in explicitly untrusted prompt blocks.
- Constrained model authority with a deterministic risk score (severity, threat intel, MITRE ATT&CK) that gates LLM calls to high-risk alerts only, a fixed JSON verdict schema with fail-safe to NEEDS_REVIEW, and an analyst review queue recording human verdicts.
- Secured the analyst chat assistant against excessive agency and data leakage: tenant-scoped MCP tools enforced server-side, topic filtering before the model runs, no disclosure of detection rules, answers grounded in tool results, and no override of stored verdicts.
- Enforced tenant isolation at four layers (mTLS certificate identity at ingress with dead-lettering of unknown clients, per-tenant OpenSearch indices, PostgreSQL RLS, app-level scoping with dual-logged cross-tenant actions); kept inference local via Ollama so no prompt data leaves the environment.
- Validated the pipeline end to end: 105 raw alerts deduplicated to ~18 incidents with 7/7 attack scenarios routed to the correct queue; built a 100–5,000 EPS load-test harness against a 100–1,000 EPS per-tenant design target.
- Performed L1–L2 incident triage, investigation and root-cause documentation across client tenants on QRadar, Seceon and Wazuh against SLAs; reported KPIs/KRIs to client stakeholders.

**Penetration Testing Intern | CFSS** **Remote | Mar 2024 – Apr 2024**

- Conducted web application vulnerability assessments with Burp Suite, Nmap and Metasploit following OWASP Top 10 and PTES methodology; documented findings with severity ratings and remediation guidance.

**Application Penetration Testing Intern | CyberHost** **Remote | Aug 2023 – Sep 2023**

- Identified application-layer flaws in the AndroGoat Android platform via manual and tool-assisted testing.

## PROJECTS

**Swaraj Hansa: Agentic AI SIEM Platform (ESDS)** | *Wazuh*, *Kafka*, *OpenSearch*, *PostgreSQL 16*, *FastAPI*, *React*, *Ollama*, *MCP*, *Docker*

- Designed tiered retention (Kafka 24h–14d, OpenSearch 30–180 days, incidents ~1 year) and a high-availability target (Kubernetes, Patroni, replicated Kafka, MinIO).

**DDoS Detection with Semi-Supervised ML** | *Python*, *network traffic analysis*

- Built a semi-supervised model combining labelled and unlabelled traffic data to detect DDoS patterns across network threat vectors (Feb 2024).

## CERTIFICATIONS

- CompTIA CySA+ (in progress)
- Microsoft Applied Skills: Design solutions with MCRA & MCSB; CAF & Well-Architected Framework
- Fortinet Certified Associate in Cybersecurity
- Google: Strategies for Cloud Security Risk Management
- Microsoft: Defender XDR Threat Protection; Zero Trust & Best Practice Frameworks
- Cyber Vidyapeeth: Cyber Defence Engineering using Python
- Udemy: SOC & SIEM (Splunk / ELK); Ansible Automation

## EDUCATION

**B.Tech Computer Science & Engineering (Cyber Security), Dayananda Sagar University, Bengaluru**
2020 – 2024 | CGPA 7.62 · CYSEC Cyber Security Club: Practical Demonstrator